

SUBRAT KUMAR JENA

Data Analyst | Data Scientist | Business Analyst | Financial Risk Analytics

Bhubaneswar (Open to Relocation) | +91 9337681466 | subratjena2911@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFILE

Engineering graduate with a habit of stress-testing answers, not just producing them. Built solutions spanning 60GB+ SEC filings, financial-risk prediction, cashflow forecasting, NLP, and business analytics, while uncovering data leakage that made a model look better than it was. Strong in Python, SQL, EDA, feature engineering, and validation, with a focus on finding flaws early and building evidence-backed solutions.

CORE SKILLS

- **Programming & Databases:** Python (Pandas, NumPy, Scikit-learn, XGBoost, CatBoost, Matplotlib, Seaborn), SQL
- **Machine Learning & AI:** SHAP, Feature Engineering, Model Evaluation, Explainable AI, Generative AI, RAG, Document Retrieval
- **Data Engineering & Analytics:** Polars, DuckDB, Parquet, ETL, Data Preprocessing, EDA, Data Validation, Tableau, Power BI, Advanced Excel
- **Backend & Tools:** FastAPI, REST APIs, Streamlit, Git, GitHub, GitHub Actions, Render, Vercel
- **Soft Skills:** analytical writing, documentation accuracy, organization, problem solving, collaboration, communication & presentation.

INTERNSHIP EXPERIENCE

Data Analyst Intern - Central Tool Room & Training Centre (CTTC), Bhubaneswar

(Jun 2024 - Jul 2024)

- Cleaned, queried, and validated structured datasets using SQL, Python and Advanced Excel to improve reporting accuracy and data consistency.
- Developed Tableau dashboards and Excel reports, transforming raw datasets into actionable business insights for analytical review.
- Performed exploratory data analysis, preprocessing, and data quality checks to support reliable reporting and structured decision-making.

Tool Design Intern - Tata Sikorsky Aerospace Limited, Hyderabad

(Jun 2022 - June 2023)

- Prepared engineering documentation for aerospace fixture workflows using Siemens NX, CATIA, and SigmaNEST in an on-site manufacturing environment.
- Optimized sheet-metal nesting and material usage, **contributing to 15%** reduction in raw material wastage across assigned workflows.
- Managed stakeholder communication across Engineering, Quality, and Production teams to align documentation and resolve workflow discrepancies.

PROJECTS

Financial Distress Monitoring Dashboard | Python, Polars, XGBoost, Streamlit, SHAP | [GitHub](#)

- **Engineered a 60GB+ SEC EDGAR data pipeline** using Polars and columnar storage, achieving **5x faster processing** than spreadsheet-based workflows for scalable financial-risk analysis.
- Built an XGBoost financial distress prediction engine with **SHAP explainability**, achieving **93% accuracy**, **0.94 weighted F1**, and **0.87 recall** on the minority distressed class.
- Integrated a **Forensic RAG Intelligence layer** to retrieve financial evidence and generate grounded explanations for predicted distress, complementing **SHAP-based model interpretation** in the Streamlit dashboard.

Cashflow Forecasting & Risk Simulation Platform | Python, FastAPI, Vercel, Supabase, Prophet, Render | [GitHub](#)

- Designed and developed an end-to-end decision-support platform with **5+ integrated modules** for cashflow forecasting, risk simulation, financial health monitoring, and dashboards.
- Automated nightly data refresh via GitHub Actions, with **SHA-256** data lineage and UUID-based async polling for reliable, verifiable pipeline runs.
- Built a conversational AI financial briefing module using **Groq Llama 3.3-70B**, FastAPI, and React, with edge-tts lingual (English) voice output for accessible financial guidance.
- Deployed the system across **2 platforms**, Render and Vercel, and iterated through debugging, API validation, and frontend-backend integration

Credit Risk Analytics Pipeline | Python, CatBoost, Scikit-learn, XGBoost | [GitHub](#)

- Resolved a pre-split oversampling leakage issue and **benchmarked 11 model configurations** across **100 random states**, with Tuned CatBoost selected as champion model at **86.18% leakage-proof accuracy** (stable & explainable, GridSearchCV).
- Reduced QA overhead by **60%** through reusable validation checks, documented preprocessing decisions and GridSearchCV-based tuning.
- Achieved minority-class (rejected loan) **F1 0.71** and approved-class **F1 0.90 post-leakage fix**; selected **CatBoost** as champion over higher accuracy **GaussianNB (87.80%)** prioritizing stability and explainability on unseen data over peak accuracy.

EDUCATION

- B.Tech, CSE (AI), Gandhi Institute For Technology, Bhubaneswar | **CGPA: 7.9** (Graduated: July 2026)
- **Relevant Coursework:** Machine Learning, Database Management, Statistics & Probability
- Post Graduate Program in Data Science, Imarticus Learning Pvt. Ltd., Hyderabad (Completed: June 2026)
- **Relevant Coursework:** Emphasis on end-to-end analytics pipeline design, KPI dashboard reporting, and stakeholder-facing data communication across Python, SQL, Excel, Tableau, and Power BI.

CERTIFICATIONS & ACHIEVEMENTS

- [Google Data Analytics](#) | [Tableau Business Intelligence Analyst \(Salesforce\)](#) | [HackerRank SQL Intermediate](#) | [LeetCode SQL 50](#)
- [ET GenAI Hackathon 2026 \(Semi-Finalist\)](#) : Built Veri-Vigil AI, a Chrome Extension misinformation detector using LLaMA 3.1, Groq, and FastAPI; selected among top teams nationally out of hundreds of submissions.
- Won a Cash Prize at [Trithon 2023](#) for exceptional problem-solving and technical collaboration skills.
- Secured a **Top 5% Merit rank** (800 out of 16,000+ candidates) in **OJEE 2023**.
- Presented E-Waste upcycling research on [Biodegradable 3D Printing Filaments](#) at RAECC-2025 National Conference.